

Substance identification based on transmission THz spectra using library search



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HIGHLIGHTS

- With renewed interest in the THz region, library search algorithms were tested.
- Spectra from the publicly viewable online RIKEN THz database were used.
- Spectral pre-processing was required for artefact removal prior to data comparison.
- Successful removal of atmospheric water vapour rotational lines is demonstrated.
- Fourier filtering for removal of channel spectra leads to efficient library search.

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ABSTRACT

Over recent years terahertz spectroscopy has become a new tool for the characterization of solid materials, in particular for investigating polymorphism and crystallinity in pharmaceuticals. Search strategies have been tested for THz spectra of various organic compounds with their spectra taken from the Riken database (<http://www.riken.jp>), using the GRAMS spectroscopy software. A subset of the entire database was used, which had been processed by removal of atmospheric water vapour lines and smoothing applied based on Savitzky–Golay convolution or time domain filtering. The spectral range available for all library substances was restricted to an interval from 0.9 to 4.5 THz (30–150 cm^{−1}). The number of vibrational bands within this spectral range is much reduced compared with mid-infrared or Raman spectra. The appropriateness of spectral pre-treatment is demonstrated with regard to reliability and robustness of the search methods. In particular, time-domain filters for smoothing and pre-treatment by the removal of water lines and etalon effects have been successfully tested in combination with least squares and correlation methods. With these insights, applications for substance identification, especially for the pharmaceutical industry, may be enlarged.

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1. Introduction

There is ever increasing demand for robust computer-based detection algorithms to enable unknown substance identification from reference digital library spectra when the composition of the samples is entirely unknown or can only be guessed. In the past, significant progress has been achieved using computer-based detection and identification based on Raman, IR, NMR and other spectral libraries. Under laboratory conditions usually a satisfactory standardized sample preparation can be established, where the conditions can be adjusted to the measurement requirements. These steps lead to high-quality spectra which are recommended to be used for establishing spectral databases. In contrast, infrared spectra measured under non-ideal and more realistic conditions

often suffer from spurious features such as atmospheric bands, etalon effects and other noise problems. The same statement applies to far-infrared (FIR) spectra, nowadays also known as THz domain. These effects complicate an automatic detection, despite the fact that modern mathematical techniques such as pattern recognition tools and elaborate library search algorithms are available, e.g. [1–4]. Previously, molecular spectroscopy with IR and Raman played an important part due to the availability of large and versatile spectral libraries. Exemplary methods employ, e.g., neural networks for the identification of polymers based on NIR-spectroscopy [5], or another recent application is the identification of food spoilage bacteria through Raman micro-spectroscopy using support vector machines [6].

Recently, there has been a regained interest for the far-infrared region or synonymously for THz spectroscopy due to novel radiation and detector technologies, especially within the pharmaceutical industry or for material science applications and

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non-destructive testing [7]. Solid formulations are the most important pharmaceutical dosage forms today. While Raman and infrared spectroscopy probe the solid state predominantly on an intramolecular level, with advantages also for amorphous substances, THz spectroscopy can probe low energy torsion and hydrogen bonding vibrations as well as the intermolecular state, i.e. long-range crystalline lattice vibrations (phonon modes). These properties make THz spectroscopy an ideal tool to investigate crystallinity and to discriminate between polymorphic forms [8]. Detection and quantification of solid phase transformations (appearance of new polymorphs) during processing is another application in pharmaceuticals.

Further examples of applications of THz technology can be found, in bio-engineering and biomedicine [9] as well as in the security field as related to explosives and hazardous materials. Several text books have been published covering the different areas [10,11]. Recently, a review on the measurement of THz spectra and chemometrics used for spectral pre-processing, qualitative and quantitative analysis has also been published [12].

This work was dedicated to testing search strategies for THz spectra of various organic compounds using the GRAMS Spectroscopy Software (<http://www.thermoscientific.com/grams>). Similar software is also available from other companies, e.g., the IDENT program from Bruker Optics (Ettlingen, Germany) using spectral library search tools or cluster analysis for the identification of chemicals. The identification of drugs and explosives from scanning persons or postal items by THz radiation was our main interest, and in this context we tested the effectiveness and performance of library search algorithms for this spectral range.

2. THz spectroscopy and spectral databases

The THz frequency range has been broadly defined from 0.1 to 30 THz. The corresponding wavelengths range from 3 mm to 10 μm and in wavenumbers from 3.33 cm^{-1} to 1000 cm^{-1} . Terahertz spectroscopy shows similar optical properties of the in wavelength neighbouring mid-infrared spectral region, and over and above, the same tools for spectral interpretation and band assignment can be exploited. Furthermore, THz radiation provides high penetration depths in special materials and low scattering combined with good spatial resolution. Due to the low energy photons, it does not induce any changes in chemical structures and, unlike UV or X-ray, THz radiation is safe for humans. THz photons lead to low-energy excitations in electronic materials, low frequency vibrational modes of condensed phase media and rotational transitions in gaseous small molecules [10]. In many ways, modern THz techniques, especially Time Domain spectroscopy (TDS), can be seen as a convenient, complementary tool to extend the possibilities for investigating the properties of many materials as previously studied by conventional Fourier transform spectroscopy.

Spectral range limitations observed for TDS spectroscopy are often up to 3 THz (100 cm^{-1}). Usually, above that upper limit, conventional FT-FIR spectrometers offer a signal-to-noise ratio equal or better than the currently available TDS instrumentation. An assessment of FT-FIR technology with its much broader spectral ranges by studying its suitability for process pharmaceutical technology has recently been published by Wu et al. [13], who tested instrumental and composition effects on terahertz spectral data quality of pharmaceutical substances.

In Fig. 1, exemplary THz spectra as obtained by the two mentioned measurement techniques are shown, which describe their different qualities. In part A, several spectra of substances embedded in a polyethylene (PE) pellet are presented with a scale from 0 to 100 cm^{-1} (0–3 THz), explicating the accessible spectral range possible with TDS under the available experimental conditions

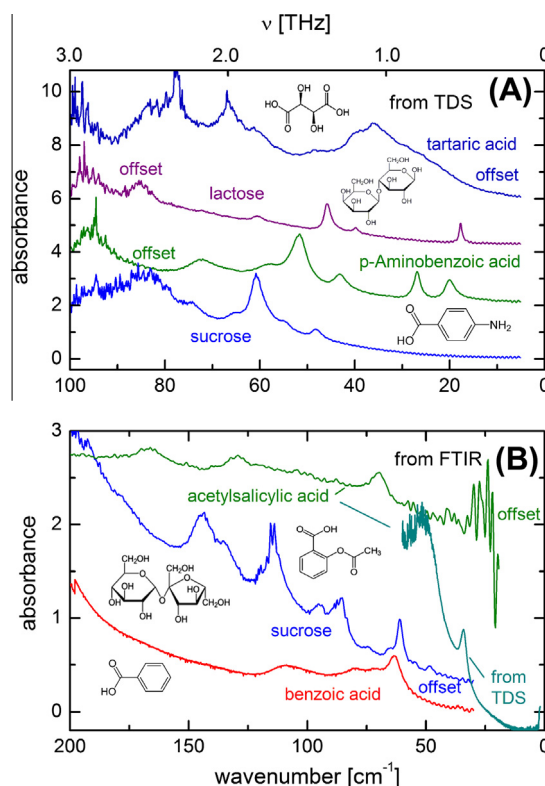


Fig. 1. Comparison of THz spectra as measured by the TDS technique (A) and by conventional FTIR-spectroscopy (B); for clarity, spectra were offset along the ordinate axis, which has also been applied to the other diagrams.

(details can be found in Ref. [14]). In the reader's interest, we also provided a THz scale to the upper diagram. The maximum interval wavenumber is evidently $\sim 100\text{ cm}^{-1}$ ($\sim 3\text{ THz}$) due to an increasing noise level. In Part B spectra from the RIKEN THz Database (more details are given below) are shown, which contain a high noise level below 30 cm^{-1} because of the marginal intensities that exist for the chosen FTIR-spectrometer with a beam-splitter selected to enable measuring a rather broad spectral range above the low interval wavenumber mentioned. In this context, it is interesting to compare the corresponding spectra of sucrose and acetylsalicylic acid measured by both techniques.

It should be noted that compared to infrared or Raman spectral libraries, only a few databases are available for the THz spectral range so far. One publicly viewable online database is the Riken THz database 2.0 [15], which has been created and is still maintained by the Tera-Photonics Team, RIKEN Sendai in Japan (<http://www.riken.jp/THzdatabase/>). At present, the database contains more than 200 spectra from the following categories: inorganic materials, agrochemicals, bio-chemical and pharmaceutical substances including saccharides, and polymers. The substances were either measured as pellets in transmission, mostly with polyethylene as matrix, as thin layers or as a mull prepared with Apiezon M. In 2007, the RIKEN THz Database and that from the Japanese National Institute of Information and Communications Technology (NICT) were merged, so that about 500 spectra were available, even though of different quality and with many redundant spectral datasets in their library (see also <http://thzdb.org>). A few spectra can also be found from the NIST webbook (<http://webbook.nist.gov/chemistry/thz-ir/>).

2.1. THz database for library search and test spectra

Our work is based on the THz spectra from the Riken database. We used a subset of the entire database, which had also been

pre-processed by removal of atmospheric water vapour lines and by Savitzky-Golay smoothing. Since the frequency range varies for the different substances, the spectral range was restricted from 0.9 to 4.5 THz ($30\text{--}150\text{ cm}^{-1}$), which was available for all substances as studied by us. The corresponding wavelengths range from $333\text{ }\mu\text{m}$ to $66.7\text{ }\mu\text{m}$. The appropriateness of spectral pre-treatment is demonstrated with regard to reliability and robustness of the search methods. In particular, Fourier filters [16], i.e. using Fourier transformation, low pass filter application and inverse transform, for smoothing and pre-treatment for the removal of residual water lines and fringes from multi-beam interferences (etalon fringes) have been tested.

In Fig. 2 the general transmission measurement scenario is described. The samples can be prepared as free-standing PE-pellets or, in the case of Apiezon mulls, between two transparent window plates. One outstanding advantage of THz spectroscopy is that many polymer materials exhibit a much higher transparency or penetration depth within the THz frequency range than for mid- and near-infrared radiation or visible light. Therefore, THz spectroscopy is preferable to IR or Raman spectroscopies, which are essentially thin layer or “surface-near” analysis methods. In our study we used 30 specially selected spectra from the entire database. Some of these substances can be found in Table 1 as results from the library search with first hits listed.

To test the performance of the library search algorithms, spectra were generated with typical interfering features found in day-to-day THz analyses or rather chosen from the raw spectral data set. In Fig. 3 several spectral features are explicated which can be attributed to “non-substance features”. There are at least four categories of spectral features that more or less complicate the identification and in particular the performance of library search algorithms used for unknown substance elucidation:

- insufficient signal-to-noise ratio,
- atmospheric interferences due to water vapour,
- etalon effect due to parallel surfaces of a sample pellet from multi-beam interference,
- additional peaks due to pellet material absorption.

As demonstrated in Fig. 3, one or more of these effects can be found in each of the library spectra. For better comparability the spectra were ordinate shifted against each other. The quartz spectrum exhibits a clear etalon fringe effect with increasing amplitudes for decreasing wave numbers. For wavenumbers above 100 cm^{-1} there was no spectral data available so that the spectrum was linearly extrapolated. There are also three polymer spectra shown (PDC 187 polyethylene spectrophotometric grade, black polyethylene sheet (PE) and black low density polyethylene (LDPE)). PE PDC 187 exhibits a comparably broad peak at 70 cm^{-1} and a clear etalon effect, but with a much higher frequency and small amplitude compared to the quartz spectrum. In contrast to PDC 187, PE and LDPE do not tend to produce significant etalon fringes. PE also shows a broad peak at approximately

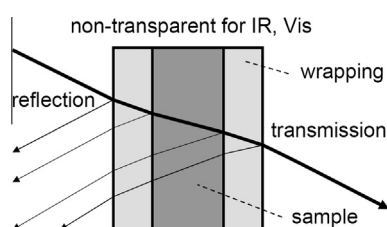


Fig. 2. Schematics of a transmission measurement scenario of substances measured as PE-pellet or wrapped in polymer films or quartz slabs.

110 cm^{-1} , whereas the LDPE spectrum is essentially flat. The lowest spectrum shows typical atmospheric line interferences due to water vapour absorption. Numerous comparably small peaks can be observed in the water vapour spectrum. These lines arise from rotational transitions of the water molecules in the atmosphere.

In the case of the water vapour rotational lines, spectra were additionally simulated using the database HITRAN96 [17] in combination with the AIRSENTRY software (Institute of Atmospheric Optics, Tomsk, Russia). The result is illustrated in Fig. 4. For better comparability, the spectra were shifted against each other. The upper diagram depicts two computed atmospheric transmittance spectra for different humidities or accordingly different path lengths. It can be seen that 50 cm path length at standard conditions ($T = 25\text{ }^{\circ}\text{C}$, $p = 1013\text{ hPa}$, $p_{\text{H}_2\text{O}} = 19\text{ hPa}$, relative humidity of 60%) is sufficient for partial saturation effects in the spectrum. In the lower diagram a simulated water vapour spectrum is compared to a measured spectrum, but shown in absorbance, which could also be used for spectral pre-processing for the removal of water vapour lines.

In Fig. 5 eight examples of the library substance spectra utilized by the authors are presented. In Fig. 5A the absorbance spectra for 2-acetylsalicylic acid, ampicillin sodium salt, fluorouracil and captan are presented. In the cases of ampicillin sodium salt and captan, the original spectra (thin lines) show spurious oscillations which are successfully removed by our Fourier filter (spectra shown as thick solid lines). In Fig. 5B, there are more example spectra, namely glycine, D-serine, trehalose and inositol after processing by means of a Fourier filter.

To test the performance of the library search algorithms the following methodology was suggested. Test contaminated substance spectra were created based on the original substance spectra from the library and realistic interferences (cf. also to Fig. 3) or unprocessed raw library spectra. In Fig. 6 the results of such a procedure and after additional Fourier filtering are shown. Here, fructose glucose, acephate and tetracene have been predominantly contaminated by means of either additional water vapour spectrum and/or etalon fringe spectra.

3. Search strategies

The performance of the search algorithms was tested based on GRAMS AI spectral ID toolbox [18]. In the first stage a library was created. For this, the spectral data format had also to be consistently matched, i.e. all points were interpolated to the same wavenumber grid and cropped to the same frequency range. The following search algorithms were tested (for the different metrics, see for example [2,3]; more details, also on the definition of the hit quality indices, are given in the Appendix):

- Euclidean distance algorithm.
- Absolute and first derivative absolute value algorithm.
- Least squares and first derivative least squares algorithm.
- Correlation and first derivative correlation algorithm.

The Euclidean distance algorithm is the industry standard algorithm for performing library data searches. The absolute value strategy does not enhance the spectral differences as much as the Euclidean distance algorithm and therefore is suitable for cases where there are additional large spikes in the unknown spectrum such as rotational water vapour lines as experienced for experimental THz-spectra. The searches using derivative spectra have the advantage of removing most of the deviations from a slowly varying baseline error in the unknown spectrum from the search results. In case the mean centering step for the correlation metrics is removed, the equation reduces to the square of the Euclidean

Table 1
Library search results (the hit-list positions are indicated by the second column, unless marked in the substance column; the correct compound is highlighted in bold face).

Method		FRUCTOSE		GLUCOSE		TETRACENE		ACEPHATE	
		HQI	Substance	HQI	Substance	HQI	Substance	HQI	Substance
Euclidean distance raw data	1	0.1502	DICOFOL	0.0903	GLUCOSE	0.1859	SILICONE	0.1426	ACEPHATE
	2	0.1592	SILICONE	0.1957	SILICONE	0.1895	ASCORB ^b	0.1553	SORBOSE
	3	... 5 ↓ ...		0.2020	FRUCTOSE	0.1895	SORBOSE	0.1871	ASCORB ^b
	4	0.1914	FRUCTOSE	0.2097	DICOFOL	0.1919	TETRACENE	0.1981	2-AC-S ACID ^a
Euclidean distance Fourier filtered	1	0.156	GLUCOSE	0.079	GLUCOSE	0.1011	TETRACENE	0.1553	ACEPHATE
	2	0.158	DICOFOL	0.206	FRUCTOSE	0.1917	SILICONE	0.1729	2-AC-S ACID ^a
	3	...	5 ↓ ...	0.212	SILICONE	0.1988	DICOFOL	0.1742	SORBOSE
	4	0.195	FRUCTOSE	0.216	HISTIDIN	0.2046	2-AC-S ACID ^a	0.1889	ASCORB ^b
Absolute value raw data	1	0.07059	FRUCTOSE	0.0431	GLUCOSE	0.0902	SILICONE	0.0902	ASCORB ^b
	2	0.07059	SUCROSE	0.0667	FRUCTOSE	0.0941	ASCORB ^b	0.0941	ACEPHATE
	3	0.0745	AMPICILLIN	0.08635	AMPICILLIN	0.1137	ACEPHATE	0.0980	SILICONE
	4	0.0745	DICOFOL	0.0941	GLYCINE	0.1176	TETRACENE	0.1020	2-AC-S ACID ^a
Absolute value Fourier filtered	1	0.067	FRUCTOSE	0.035	GLUCOSE	0.0588	TETRACENE	0.0863	ACEPHATE
	2	0.067	SUCROSE	0.067	FRUCTOSE	0.0745	DICOFOL	0.0902	2-AC-S ACID ^a
	3	0.075	AMPICILLIN	0.086	GLYCINE	0.0823	SILICONE	0.0902	ASCORB ^b
	4	0.078	GLYCINE	0.090	AMPICILLIN	0.0941	TETRALOSE	0.0980	SILICONE
Least squares raw data	1	0.0083	DICOFOL	0.0028	GLUCOSE	0.0139	ASCORB ^b	0.0119	ACEPHATE
	2	0.0090	FRUCTOSE	0.0094	FRUCTOSE	0.0149	SILICONE	0.0137	ASCORB ^b
	3	0.0099	SUCROSE	0.0152	AMPICILLIN	0.0182	TETRACENE	0.0141	SILICONE
	4	0.0100	AMPICILLIN	0.0182	INOSITOL	0.0201	ACEPHATE	0.0146	2-AC-S ACID ^a
Least squares Fourier filtered	1	0.009	FRUCTOSE	0.0022	GLUCOSE	0.0063	TETRACENE	0.0108	ACEPHATE
	2	0.009	AMPICILLIN	0.0096	FRUCTOSE	0.0134	SILICONE	0.0115	2-AC-S ACID ^a
	3	0.010	DICOFOL	0.0154	AMPICILIN	0.0155	2-AC-S ACID ^a	0.0138	ASCORB ^b
	4	0.010	SUCROSE	0.1803	INOSITOL	0.0163	DICOFOL	0.0167	SILICONE
1st Derivative least squares raw data	1	3.1E-05	SERINE	6.2E-05	GLUCOSE	0.0007	MANNOSE	0.0005	DICOFOL
	2	3.1E-05	FRUCTOSE	7.7E-05	D-SERINE	0.0008	TETRACENE	0.0005	2-AC-S ACID ^a
	3	3.1E-05	SILICONE	7.7E-05	FRUCTOSE	0.0008	SORBOSE	... 33 ↓ ...	
	4	4.6E-05	2-AC-S ACID ^a	7.7E-05	SILICONE	0.0009	2-AC-S ACID ^a	0.0010	ACEPHATE
1st Derivative least squares Fourier filtered	1	1.5E-05	D-SERINE	1.5E-05	GLUCOSE	3.1E-05	2-AC-S ACID ^a	5.1E-05	2-AC-S ACID ^a
	2	1.5E-05	FRUCTOSE	3.1E-05	2-AC-S ACID ^a	3.1E-05	D-ALANINE	5.1E-05	AMPICILIN
	3	1.5E-05	SILICON	3.1E-05	ASPARTAM	... 23 ↓ ...		... 33 ↓ ...	
	4	3.1E-05	2-AC-S ACID ^a	3.1E-05	D-ALANINE	6.2E-05	TETRACENE	3.4E-04	ACEPHATE
Correlation raw data	1	0.0352	SILICONE	0.0081	GLUCOSE	0.0208	TETRACENE	0.0487	2-AC-S ACID ^a
	2	0.0352	GLUCOSE	0.0625	SILICONE	0.0681	L-ALANINE	0.0626	ACEPHATE
	3	... 6 ↓ ...		0.0683	FRUCTOSE	0.0716	D-ALANIN	0.0766	SORBOSE
	4	0.0600	FRUCTOSE	0.0694	HISTIDIN	0.0750	SILICONE	0.0951	ASCORB ^b
Correlation Fourier filtered	1	0.034	SILICONE	0.007	GLUCOSE	0.0181	TETRACENE	0.0450	2-AC-S ACID ^a
	2	0.035	GLUCOSE	0.060	SILICONE	0.0642	L-ALANINE	0.0580	ACEPHATE
	3	... 8 ↓ ...		0.0671	FRUCTOSE	0.0678	D-ALANINE	0.7306	SORBOSE
	4	0.061	FRUCTOSE	0.068	HISTIDIN	0.0714	SILICONE	0.0911	ASCORB ^b
1st Derivative correlation raw data	1	0.4737	FRUCTOSE	0.5217	GLUCOSE	0.5454	MANNOSE	0.7273	MANNOSE
	2	0.5789	CAPTAN	0.6087	FRUCTOSE	0.6364	TETRACENE	0.7500	DICOFOL
	3	0.6316	ASPARTIC ACID	0.6522	CAPTAN	0.6727	SORBOSE	... >30 ↓ ...	
	4	0.6316	GLUCOSE	0.6522	DICOFOL	0.8000	DICOFOL	0.8409	ACEPHATE
1st Derivative correlation Fourier filtered	1	0.500	FRUCTOSE	0.400	GLUCOSE	0.4761	TETRACENE	0.4500	FLUORO ^c
	2	0.600	TETRACENE	0.550	FRUCTOSE	0.4761	CAPTAN	0.5499	2-AC-S ACID ^a
	3	0.600	GLUCOSE	0.600	ASPARTAM	0.5238	FLUORO ^c	... 27 ↓ ...	
	4	0.650	ASCORB ^b	0.600	CAPTAN	0.6190	BENZOIC ACID	0.8000	ACEPHATE

^a 2-Acetylsalicylic acid.

^b Ascorbic acid.

^c Fluorouracil.

distance. In almost all cases, correlation searching will provide better or equal hit quality information when compared to Euclidean distance searching.

4. Results and discussion

Exemplary spectra of the Riken database representing some of the main substance classes are shown in Fig. 5. The spectral range available differs significantly for the individual substances within the library, as some spectra were recorded for the interval of 30–300 cm⁻¹ with quite different number of data points (for example, glucose spectrum 790, ampicillin 630 or captan 7500 data points). For captan and ampicillin 150 data points were taken into account in the Fourier domain for the Fourier filter for removing etalon fringes effectively. An experimental problem is water vapour

absorption, when only non-evacuated spectrometers are available. The experimental spectra have shown good agreement with those simulated using the HITRAN data base and our AIRSENTRY software (see also Fig. 4).

Using the different sample spectra, which are displayed in Fig. 6, library searches were carried out using the different metrics as discussed above. Results have been listed in Table 1. Here we found that the 1st derivative absolute value method was not useful at all, since the HQIs had been extremely similar and only the “raw” glucose spectrum was listed as no. 3 in the best case; in all other cases the position was between 10 and 33. Therefore, no results have been listed for this search strategy.

In principle, it can be stated that the Fourier filtering as pre-processing improved the library search results as expected. Here, the Euclidean, the absolute value, least squares and correlation methods performed similarly with regard to the position of the

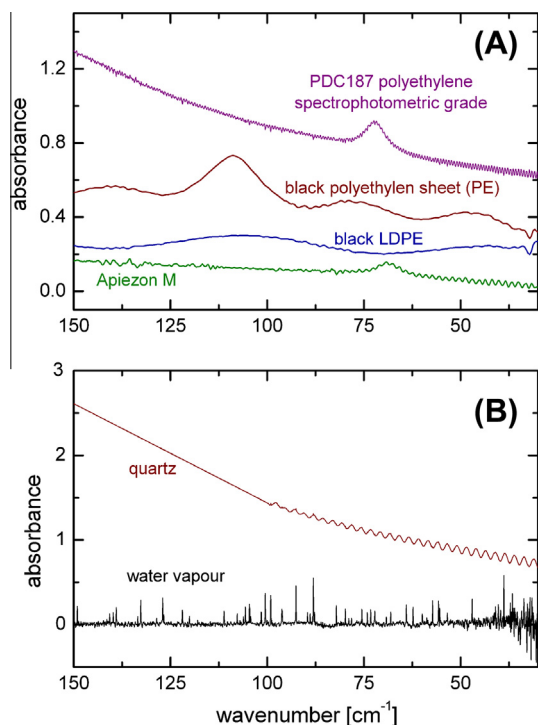


Fig. 3. Different interferents added to raw spectral data as used for library search and testing: A Embedding substances used for pellet and mull preparations, exemplary quartz substrate spectrum and water vapour rotational spectrum as obtained from Riken library spectra.

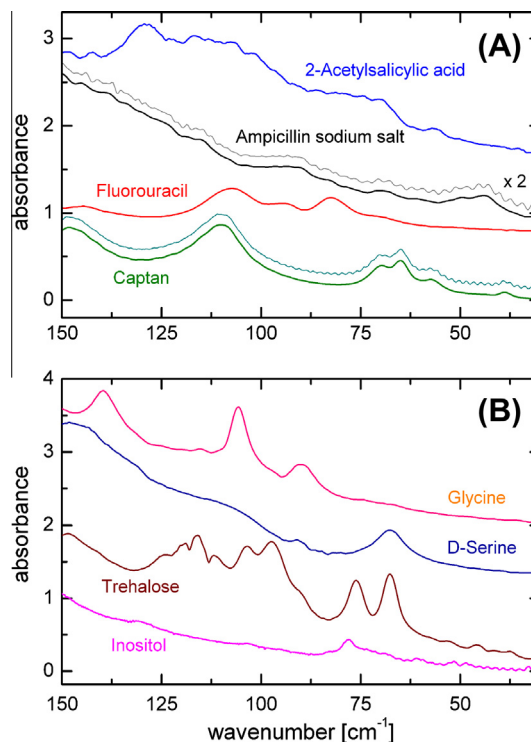


Fig. 5. Examples of ‘raw’ Riken library spectra and after processing; for water vapour contaminated spectra, see also Fig. 6B.

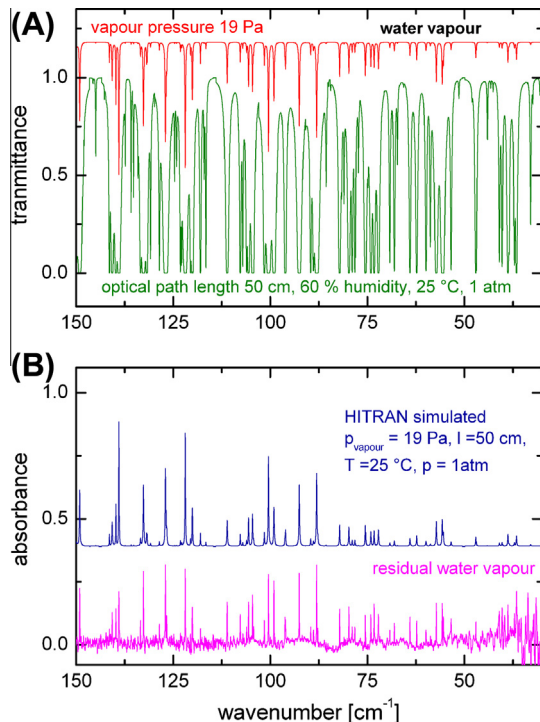


Fig. 4. Spectra illustrating atmospheric absorption: (A) Transmission spectra (lower trace: optical path length 50 cm, 60 % rel. humidity (19 hPa), 25 °C, total pressure 1013 hPa, upper trace: water vapour pressure 19 Pa); (B) simulated and experimental water vapour absorbance spectra.

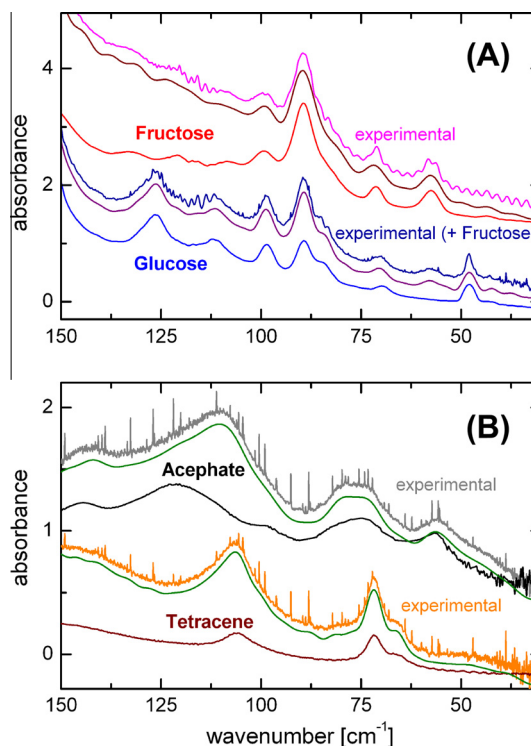


Fig. 6. Alienated substance spectra (experimental and Fourier-filtered slightly off-set) compared with the corresponding library data.

“unknown” spectrum in the hit-list. For the 1st derivative algorithm with least squares and correlation, the results for acephate

were poor (positions 33 and 27), whereas for tetracene correlation produced a first place in the hit-list. For the latter compound, least squares resulted in a no. 23 position, although with “raw” data we

reached second place in the hit-list. For the glucose spectrum, which had some fructose admixed, we always found the fructose spectrum as a search result due to spectral similarity. A general problem is the sloping baseline found in all spectra including the library entries. There is the option of a linear baseline correction in GRAMS Spectroscopy Software, but this is applied to the sample spectrum only, so that differences in baselines can deteriorate the search results significantly, as this influences the similarity metrics.

5. Conclusions

In this paper conventional library search algorithms were tested for THz spectra. THz or FIR spectroscopy provides several unique attractive features that make this frequency range for some measurement tasks preferable to other optical spectroscopy. However, it can be stated that visual inspection is still a step, which is recommended for selecting the correct library entry from the suggested hit list spectra obtained by library searching. In the recent past, in particular so-called THz TDS (time domain spectroscopy) devices have become popular and will be found more widely distributed in the future with instrumentation becoming less expensive. Exploring THz spectroscopy also as an alternative technique for directly characterizing, e.g., pharmaceutical materials will provide insight on adapting this spectroscopic method as a potential process analytical tool. In such applications chemical identification by library search is important for checking substances as starting compounds, final products or when raw or refined products enter the factory premises. This will be certainly in competition with infrared, near-infrared and Raman spectroscopy.

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Appendix A

The definitions of the various **HQIs** obtained by spectral library search are as follows [18]:

1. Euclidean Distance Algorithm

The Euclidean distance of an unknown to a library entry is effectively a normalized least squares dot product on the unknown: The Hit Quality Index (**HQI**) is calculated from the following formula:

$$HQI = \sqrt{2} \cdot \sqrt{1 - \frac{A_R \cdot A_S}{\sqrt{A_R \cdot A_R} \sqrt{A_S \cdot A_S}}} \quad (1)$$

where A_R is the library entry being searched and A_S is the unknown spectrum with n number of data points. This formula gives the best match at distance of 0 ($A_S = A_R$) and the worst at distance of $\sqrt{2}$. Spectral regions to be excluded from the calculation are obtained by setting the corresponding values in the two vectors to zero.

2. Absolute Value Algorithm

The absolute value **HQI** of the unknown from a library entry is calculated as the sum of the absolute difference vector:

$$HQI = \frac{\sum_{i=1}^n |A_{S,i} - A_{R,i}|}{n} \quad (2)$$

Since all absorbance spectra have been normalized with the minimum set to 0 and the maximum to 1, the best match is 0 and the worst is 1. Note that this algorithm provides poor discrimination among hits due to very similar **HQI** values.

3. Least Squares Algorithm

The least squares **HQI** of the unknown from the library entry are calculated as the sum of the squares of the vector difference:

$$HQI = \frac{\sum_{i=1}^n (A_{S,i} - A_{R,i})^2}{n} \quad (3)$$

The same **HQI** range is valid as for Eq. (2).

4. First Derivative Absolute Value and First Derivative Least Squares Algorithms

Before the Absolute Value or Least Squares **HQI** is calculated, the derivatives of the unknown and library spectrum are computed. This derivative is simply the difference between each set of points in the spectrum. For the 1st derivative absolute value, the **HQI** reads

$$HQI = \frac{\sum_{i=1}^n |(A_{R,i} - A_{R,i-1}) - (A_{S,i} - A_{S,i-1})|}{n} \quad (4)$$

Similarly, the 1st derivative least squares **HQI** of the unknown from the library entry is calculated as the sum of the squares of the 1st derivative vector difference.

5. Spectrum Correlation and 1st Derivative Correlation Algorithm

This algorithm is very similar to the Euclidean distance algorithm with one very important difference: both the unknown and the library data are mean-centered, which gives $\bar{A}_R$ and $\bar{A}_S$, before the vector dot products are calculated:

$$HQI = 1 - \frac{(\bar{A}_R \cdot \bar{A}_S)^2}{(\bar{A}_R \cdot \bar{A}_R)(\bar{A}_S \cdot \bar{A}_S)} \quad (5)$$

When the signal to noise ratio and baselines in the library and unknown spectra are very similar, the mean values will be effectively the same. The 1st derivative correlation algorithm is the same as for absorbance spectra, except that the first derivative of both the unknown and library spectra are taken before calculating the **HQI**.

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